

MindfulBreath: Every Breath You Take, Every Mind You Wake

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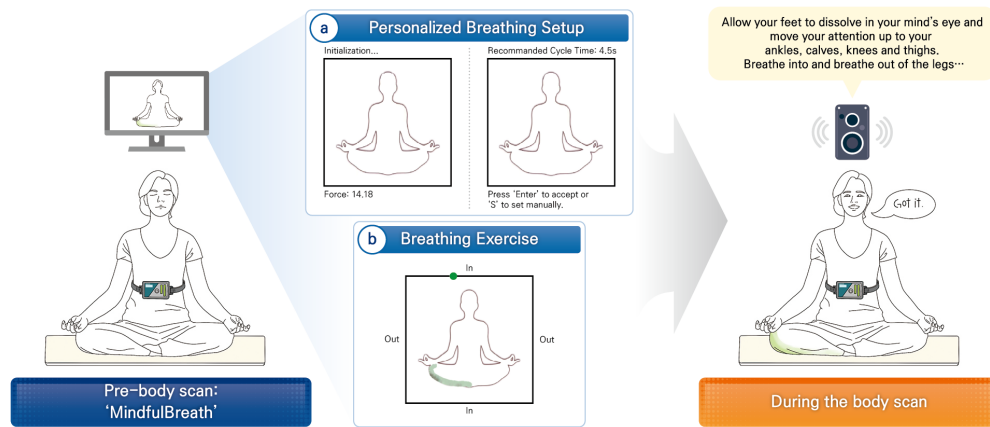


Figure 1: Overview of the MindfulBreath system. (a) Personalized Breathing Setup: The system analyzes natural breathing patterns to suggest a personalized rhythm. **(b) Breathing Exercise:** Visual feedback synchronized with breath helps users focus and prepare for deeper body scan.

Abstract

Breathing is a common entry point to mindfulness. But for beginners, simply observing the breath can feel vague and hard to follow. We present *MindfulBreath*, an interactive system that uses real-time respiratory data to recommend a personalized breathing pace and visualize each inhale and exhale as flowing movement

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through a human figure. Designed as a preparatory experience, MindfulBreath helps users rehearse attentional control and embodied focus before beginning body scan meditation. In a study with 15 participants, Initial findings suggest that MindfulBreath may improve focus, relaxation, and engagement over passive music or standard breathing apps. These preliminary results indicate the potential visually embodied breathing exercises effectively bridge to deeper mindfulness. Unlike conventional mindfulness apps and devices, which offer fixed timing or generic animations, MindfulBreath adapts in real time to each user's unique respiratory pattern, positioning breath not only as a target of observation, but as an accessible medium for embodied mental training.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Mindfulness; Breathing exercise; Biofeedback; Respiratory sensing; Interactive system; Human-centered computing

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1 Introduction

Mindfulness gains wide recognition for promoting mental well-being by supporting present-moment awareness and emotional regulation [11, 24]. Among its core techniques, body scan meditation is often used to cultivate sustained attention to internal sensations. However, for beginners, this practice can feel cognitively demanding and difficult to understand [10, 14]. Supporting this transition, instructors often start with breath-based guidance, as breathing is ever-present and controllable. Yet common approaches rely on abstract cues like "pay attention to your breath" with little structure. Without concrete support, beginners often lose focus, making it harder to transition into body scan practices.

To address this gap, we developed **MindfulBreath**, a preparatory system that helps mindfulness novices cultivate attentional readiness. The system captures real-time respiratory signals and provides personalized pacing, visualized through dynamic breath flows across regions of a seated human figure aligned with body scan meditation. This structured routine offers a grounded way to shift attention inward and stabilize focus before body scan. Our evaluation is guided by the following research questions:

- **RQ1.** How does preparatory breathing affect physiological and attentional states during mindfulness, compared to non-breathing activities?
- **RQ2.** How does personalized visual biofeedback enhance mindfulness engagement, compared to a generic breathing app?

Through this work, we aim to position breath not as a vague object of observation, but as a concrete interface for embodied mental training. The source code and visual materials will be made publicly available upon publication at <https://github.com/ohchannah/MindfulBreath>

2 Related Work

To support novices in mindfulness, prior work has explored technologies that make internal experiences more tangible. Multimodal feedback—through visual, auditory, or tactile channels—has been used to scaffold attention and improve user engagement [1, 25, 28]. Breathing, as a naturally accessible and controllable process, has served as a core interaction modality in such systems. Interactive applications and games visualize breath or react to respiratory input, encouraging users to regulate pace and reduce stress [3, 18, 21, 23]. Some systems aim to sustain attention through immersive audiovisual cues [17], while others use tangible feedback to support calm and embodiment [16, 29]. However, most focus on support during

meditation itself. Few consider how breathing might help users prepare for cognitively demanding practices like body scan. To address this, we treat breath as a rehearsal tool that guides attention and helps beginners settle into mindfulness.

3 Methodology

MindfulBreath consists of three stages: Initialization, Breath Cycle Recommendation, and Breathing Exercise (Figure 1). **Initialization.** Users begin with a 30s calibration where a wearable sensor records respiratory force. This data is normalized to establish personalized minimum and maximum breath levels, accounting for differences in breath strength and pace. **Breath Cycle Recommendation.** After initialization, the system analyzes the user's natural breathing rhythm and recommends a slightly prolonged pace—roughly 10% slower than their average cycle. The system adjusts to the user's natural rhythm, easing transition into slower breathing [2]. This helps avoid the discomfort or disengagement often caused by rigid, one-size-fits-all pacing. **Breathing Exercise.** This stage delivers a breathing exercise—not a meditative practice. Rather than observing the breath passively, users actively regulate their breathing according to a guided rhythm. To support pacing, a dot continuously moves along the four edges of a square path, where each side represents either inhalation or exhalation—completing one full loop every two breath cycles. Simultaneously, real-time respiratory force data drives flowing lines that animate through different parts of a seated human figure. These lines progress in the same sequence as a traditional body scan—spine, right arm, left arm, head, right leg, left leg, and heart—shifting to a new body region with each completed square loop. This dual visual system links sensed breath with bodily progression, helping users internalize a slower rhythm and orient attention toward embodied awareness in preparation for body scan practice.

3.1 Experiment Design

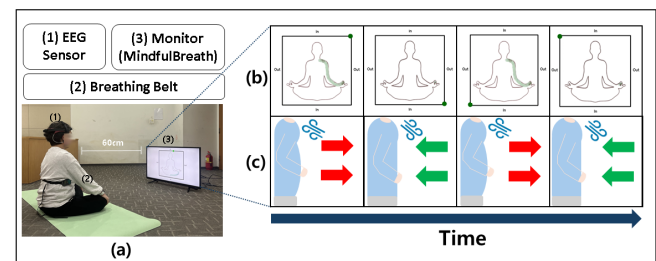


Figure 2: (a) Experimental setup and environment. (b) Animation of MindfulBreath during two full breathing cycles. (c) Corresponding visualization of the user's inhalation and exhalation.

To evaluate the proposed MindfulBreath program, we compared various preparatory activities and their effects on mindfulness during body scan meditation—a fundamental practice in focused attention meditation [15]. This study aims to systematically investigate how meaningful preparation processes influence physiological states and support meditation, beyond simple tool comparison.

We recruited a total of 15 participants (6 males, 9 females; ages ranged from 19 to 30 years, $M = 25.4$, $SD = 2.95$). They were then

divided into three groups (A, B, C), with five participants in each group. Each group completed a different five-minute pre-body scan activity. Group A listened to mindfulness music to promote relaxation. Group B used a smartphone-based breathing app (*The Breathing App*¹) featuring a 4:4 inhale-to-exhale ratio guided by a 3D sphere animation. Group C performed interactive breathing exercises using our breathing program, MindfulBreath.

All participants were meditation-naïve adults who provided informed consent. The study took place in a quiet, distraction-free environment (Fig. 2a). Participants wore a breathing belt and EEG sensor, seated 60 cm from the display running the MindfulBreath program. A one-minute baseline was recorded (Group C used this time for Initialization and Breath Cycle Recommendation). Each participant then completed a five-minute preparatory activity, followed by a standardized 15-minute guided body scan meditation [27], and concluded with self-report surveys and interviews. Figures 2b–c show the animation and corresponding breath visualization.

3.2 Measures

We evaluated the interventions using self-report scales, physiological signals (EEG and respiration), and interviews. **1. EEG:** EEG data were collected using the **BIOSS S8** device, a portable 8-channel dry electrode system, focusing on frontal regions (Fp1, Fp2) linked to attention and emotion [20]. Additionally, a 50Hz notchfilter was used to suppress power line interference [5]. Preprocessing (filtering, ICA) was applied [12] to extract power for Theta (4–8 Hz), Alpha (8–13 Hz), SMR (12–15 Hz), and Mid-Beta (15–20 Hz). The (SMR + Mid-Beta)/Theta ratio was used as an attentional index [4]. **2. Respiration:** Breathing data were collected via a Vernier GDX belt. After filtering and smoothing, BPM was extracted using a sliding window (60s, 30s overlap) [9, 19]. Median BPM (6–30 range) was analyzed [26]. **3. Self-report:** Mindfulness was assessed with the State Mindfulness Scale (SMS) [22], covering awareness of body and mind, where higher scores reflect greater state mindfulness. We also used a modified Engagement and Effectiveness Survey [13], a 5-point Likert scale measuring intrinsic value, perceived effectiveness, and task engagement, where higher scores reflect greater participant engagement. **4. Interview:** Post-study interviews explored participants' subjective experiences with the preparatory tasks and their impact on body scan engagement.

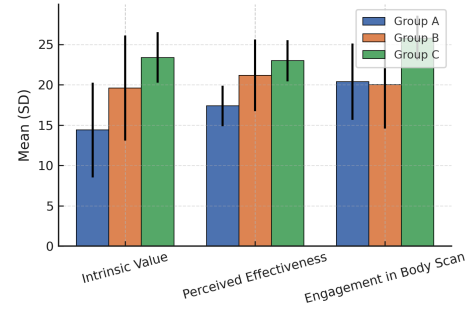
4 Results

Despite limited group size ($n=5$), we used descriptive statistics to explore potential response trends. These results should be interpreted as indicative rather than conclusive.

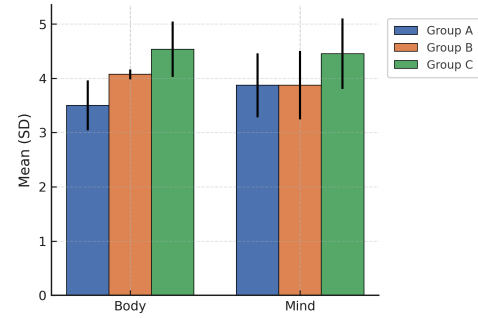
Table 1: EEG Frequency Band Metrics

Group	Alpha	Beta	Theta	(SMR+MB)/Theta
A	10.57	33.46	9.13	2.47
B	11.33	33.96	9.18	2.50
C	11.17	36.49	9.50	3.13

¹<https://apps.apple.com/us/app/the-breathing-app/id1285982210>



(a) Engagement with Preparatory Activity



(b) State Mindfulness Scale (SMS) with body scan

Figure 3: Results from the self-report surveys. All values are presented as Mean (SD).

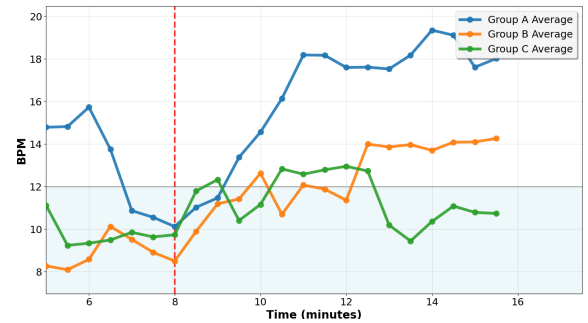


Figure 4: Respiratory rate (BPM) during body scan. Mean (range) – A 15.62 (12.86–18.98), B 12.04 (9.76–13.53), C 10.91 (8.01–15.56).

4.1 RQ1: Impact of Breathing Exercises

EEG. Group-level differences in attentional stability, measured by the (SMR + MidBeta)/Theta ratio, revealed clear effects of the preparatory activities. Group A, which only listened to music, showed the lowest ratio ($M=2.47$). In contrast, Group C (MindfulBreath) recorded the highest ($M=3.13$), followed by Group B ($M=2.50$). These results suggest that preparatory breathing exercises more effectively enhanced sustained attention than passive relaxation.

Respiration Data. Figure 4 shows that breathing exercise groups achieved 23% and 30% lower breathing rates respectively than the

control group, demonstrating the effectiveness of the breathing exercises. Distinct differences in temporal patterns were also observed between groups. Groups B and C maintained stable breathing rates throughout the measurement period, whereas Group A, which did not undergo breathing training, showed a gradual breathing rate increase in the later phase.

Self-report & Interview. Group A participants described music as relaxing but disconnected from body scan. In contrast, participants who engaged in breathing exercises (Groups B and C) reported stronger focus and engagement with the body scan. This was reflected in higher self-reported scores for both perceived effectiveness and engagement (Figure 3a). Interview data suggest that preparatory breathing helped provide a stable attentional anchor, supporting attentional anchoring throughout the meditation. While different in design, both breathing interventions offered a structured entry into mindfulness practice that music alone could not provide.

4.2 RQ2: Impact of Personalized Biofeedback

EEG. Compared to Group B, the personalized, biofeedback-driven MindfulBreath system (Group C) resulted in consistently higher Alpha and Beta band power, reflecting greater calmness and attentional engagement. Group C also recorded the highest (SMR+Mid-Beta)/Theta ratio ($M=3.13$), a marker of sustained concentration, outperforming Group B ($M=2.50$) [12].

Respiration Data. As shown in Figure 4, during the initial 8 minutes of meditation, all three groups exhibited similar breathing patterns within the 10–12 BPM range [7]. However, after the introduction of a complex cognitive task at the 8-minute mark, clear distinctions emerged. This is interpreted as reflecting differences in breathing regulation and coping strategies among the groups. Group B maintained a consistently elevated breathing rate of 14–15 BPM during the latter part of the session [8]. In contrast, Group C, influenced by personalized biofeedback training, sustained a parasympathetic-dominant state even under the same stress conditions. Although there were temporary fluctuations, Group C largely maintained a stable breathing range below 9–13 BPM. This demonstrates that personalized breathing training contributed not only to initial adaptation but also to maintaining stability during the extended body scan process [6]. While visual trends suggest differences across groups, statistical testing was not conducted due to small sample size ($n=5/\text{group}$), which limits the validity of inferential statistics.

Self-report & Interview. Interview data contextualize trends in Figure 3b, especially for Group C. While participants in Group B found the standard app's rigid 4:4 rhythm to be monotonous, interviews with Group C revealed three primary design benefits of MindfulBreath that drove higher engagement: (1) *Personalized Breathing Rhythm*: Participants noted that the system's adaptive pacing aligned well with their natural breathing patterns, making the experience more comfortable and intuitive to follow. One participant explained, "Training adapted to my lung capacity, reducing the burden.". This personalization stood in sharp contrast to the one-size-fits-all approach of the standard app. (2) *Physiological Rehearsal for Mindfulness*: Several participants noted that the system's responsive biofeedback, acted as a form of *rehearsal* for the body scan. As one participant described, "Since I saw my breathing

flow in advance before participating in meditation, when focusing on each body part, I tried to breathe while visualizing the flowing image". The breath visualization not only mirrored the flow of the upcoming meditation but also helped them maintain attentional focus without cognitive breaks. (3) *Engaging Visual Guidance*: The real-time, dynamic visualization of their own breath was found to be significantly more engaging than the passive animated sphere of the standard app. One participant captured this experience by saying, "Synchronized screen changes made me feel one with the system.". This interactive biofeedback transformed the abstract task of "focusing on breath" into a concrete and responsive visual experience, which participants found motivating. The effectiveness of these key aspects demonstrates that biosensing can transform abstract mindfulness concepts into concrete, accessible experiences for beginners.

5 Conclusion

Our exploratory study suggests that preparatory, interactive breathing exercises may meaningfully enhance engagement and concentration during a subsequent body scan session, especially for mindfulness novices. Our study highlights two key mechanisms supporting the effectiveness of preparatory breathing for mindfulness novices. Cognitively, MindfulBreath offers a rehearsal phase that stabilizes attention through personalized breath pacing. The system externalizes abstract instructions by synchronizing biofeedback with visual cues, helping beginners internalize core mindfulness actions. By lowering the conceptual barrier to entry, MindfulBreath empowers users to embody the practice rather than merely follow it. Its personalized and adaptive design tailors the experience to individual users, offering a more effective and approachable entry point into practice. Our findings support the broader vision that biosensing—especially when paired with intuitive, multi-modal feedback—can serve as a powerful medium for making mindfulness more accessible, interpretable, and engaging for a wider audience.

6 Limitations and Future Work

While MindfulBreath shows promise as a preparatory mindfulness tool, several limitations remain. Its reliance on a dedicated breathing belt constrains accessibility, motivating integration with more ubiquitous sensors. The visual feedback also prevented support during closed-eye meditation, indicating the potential of vibrotactile or auditory modalities for continuous guidance. Finally, the evaluation was limited to a single session with a small sample. Future studies should assess longer-term effects across diverse practices and participant populations.

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